



UNIVERSITÀ
DI TRENTO

Department of
Information Engineering and Computer Science

Automated Reasoning and **Formal Verification**

Laboratory 11

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<https://github.com/masinag/arfv2026>

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These slides are derived from those by Stefano Tonetta, Alberto Griggio, Silvia Tomasi, Thi Thieu Hoa Le, Alessandra Giordani, Patrick Trentin, Giuseppe Spallitta for FM lab 2005-2024.



Real time systems

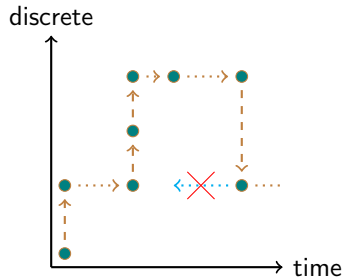
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- ▶ Safety-critical domains: defense, transportation, health-care, space, avionics, etc.

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- ▶ All clocks increase equally in time-elapses
- ▶ Model Checking for TTS is **undecidable**



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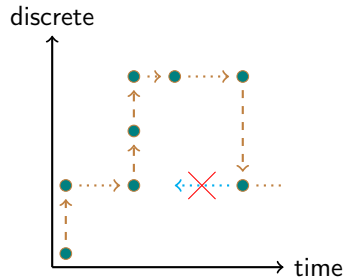
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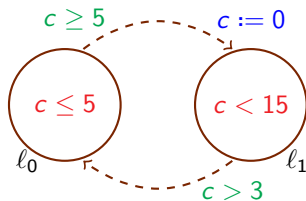
Timed Automata (TA)

- ▶ **Decidable** restriction of TTS
- ▶ **Finite Time Abstraction:**
clocks compared only to constants



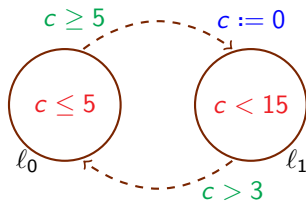
Timed Automata (TA)

- Explicit graph representation: discrete states (nodes) and transitions (edges).
- Symbolic representation of temporal aspects via (convex) constraints:
location invariants, transition guards and resets.



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location invariants, transition guards and resets.



Symbolic TTS

- ▶ Formulas represent sets of states: $p := \{s \mid s \models p\}$
- ▶ Symbolic transitions $\varphi(X, X')$
- ▶ There is a discrete transition $s_0 \rightarrow s_1$ iff $s_0(X), s_1(X') \models \varphi(X, X')$

⇒ can be described in (timed) nuXmv!

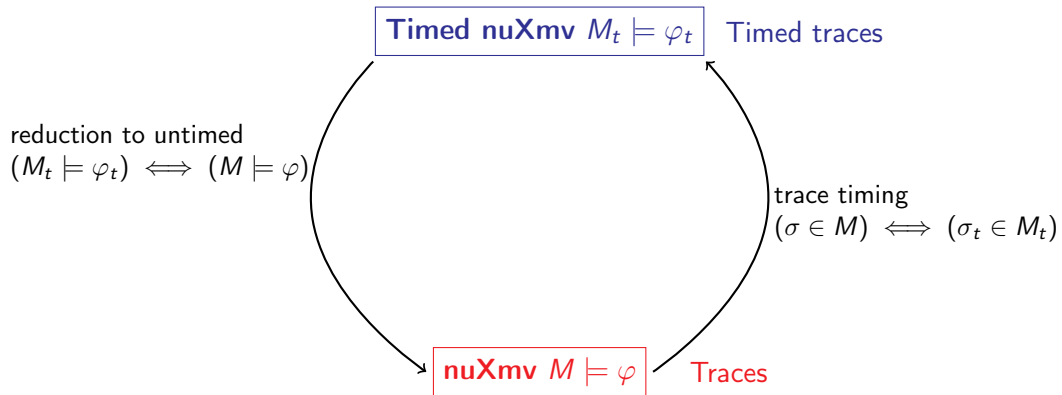
$$\begin{aligned}
 (l = \ell_0) &\rightarrow (c \leq 5) && \wedge \\
 (l = \ell_1) &\rightarrow (c < 15) && \wedge \\
 (l = \ell_1 \wedge l' = \ell_0) &\rightarrow (c > 3) && \wedge \\
 (l = \ell_0 \wedge l' = \ell_1) &\rightarrow && \\
 &((c \geq 5) \wedge (c' = 0)) &&
 \end{aligned}$$



Outline

1. Timed nuXmv
2. Timed and Infinite Traces
3. Exercises
4. Homework

nuXmv for Timed System: Architecture¹



¹ **Alessandro Cimatti et al.** "Extending nuXmv with Timed Transition Systems and Timed Temporal Properties". In: *CAV 2019*
1. Timed nuXmv



Overview

Extension of nuXmv for real-time system modelling and verification:

- ▶ Allows for modelling Timed Transition Systems with **continuous time domain**
- ▶ Enables specification and verification of **timed properties**



Time Domain and Clock Variables

- ▶ Must start with `@TIME_DOMAIN` continuous;
- ▶ New variable type: `clock`
- ▶ `clock` variables increase uniformly with time
- ▶ Built-in `clock` variable: `time`

System Specification

INIT : Specifies initial conditions

TRANS : Constrains **discrete** behavior **only**

INVAR : Defines invariant conditions. Clocks allowed in invariants with shape:
(no clock expr) -> (convex clock expr);
(convex expression: conjunction of atoms)

Timed Transitions

- ▶ System evolution alternates between:
 - ▶ **discrete** transitions: instantaneous state changes
 - ▶ **timed** transitions: passage of time where all clocks increase by the same amount
- ▶ **URGENT** conditions (time-freeze): when one of the **URGENT** conditions is satisfied, only discrete transitions are allowed
- ▶ **noncontinuous** type modifier: allows variables to change their value during timed transitions

Specifying Properties

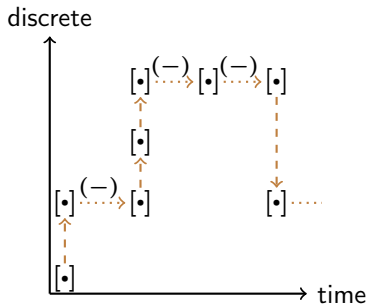
Specifications in **Metric Temporal Logic**^a ($MTL_{0,\infty}$):

- ▶ LTL operators: **X**, **Y** (yesterday) **U**, **S** (since), **F**, **G**
- ▶ Bounded LTL operators: **F[a,b]**, **G[a,b]** (finally/globally if $time' \in time + [a, b]$)
- ▶ Different operators to refer to the **discrete** next (**X**) and **timed** next (**X~**), and symmetrically for the past (**Y** and **Y~**)
- ▶ Operators to get the value of **expr** the next/last time **p** will hold/held:
time_until(p) is the time until p will hold
time_since(p) is the time since p held
expr @F~ p is the value expr will have the next time p will hold.
expr @O~ p is the value expr had the last time p held

^ahttps://en.wikipedia.org/wiki/Metric_temporal_logic

Timed to Untimed Model

- ▶ **clock** symbols and **time**: variables of type **real**
- ▶ δ : the amount of time elapse for every transition (continuous positive variable)
- ▶ ota' : prescribes the alternation of singular $[\cdot]$ and open $(-)$ time intervals



Properties Rewriting

MTL *fragment*

$$F_{[0,5]} p$$

↓ rewrite

LTL *timed*

$$((\neg p U p) \wedge \text{time_until}(p) \leq 5) \vee \\ ((\neg p U \tilde{X} p) \wedge \text{time_until}(p) < 5)$$

↓ untime

LTL *untimed*

$$((\neg p U p) \wedge (\text{time}@ \tilde{F} p) - \text{time} \leq 5) \vee \\ ((\neg p U ((\neg \iota \wedge p) \vee X(\neg \iota \wedge p))) \wedge (\text{time}@ \tilde{F} p) - \text{time} < 5)$$



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From Untimed Model Execution to Timed Trace

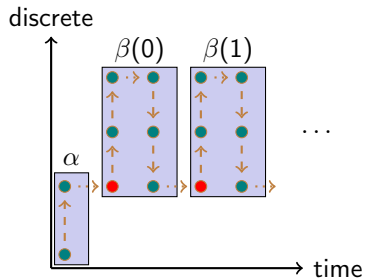
Issue

- ▶ nuXmv traces must have **lasso shape**: $\alpha\beta^\omega$, with α and β sequences of states.
 $\implies$ complete for finite state systems.
- ▶ In **timed** and **infinite** state systems they may not exist!

Solution

Lasso-shaped traces with **diverging variables** $\alpha\beta(i)^\omega$:

- ▶ Some variables **repeat** (like in lasso-shaped traces)
- ▶ Other variables **diverge** as a function of previous state:
 $s_i(v) = f_v(s_{i-1})$
e.g. $next(time) := time + \delta$





How to Run: Model [1/3]

`./nuXmv -time -int` start nuXmv interactively and enable commands for timed models.

`go_time` process the model.

`write_untimed_model` dump SMV model corresponding to the input timed system.



How to Run: Verify [2/3]

`timed_check_invar` check invariants

`timed_check_ltlspec` check LTL

Mostly the same command line options of the corresponding commands for untimed models.



How to Run: Simulation and Traces [3/3]

`timed_pick_state` pick initial state.

`timed_simulate` simulate the model starting from a given state.

Formally nuXmv uses a **super-dense** **weakly-monotonic** time model $T \subset \mathbb{N} \times \mathbb{R}_0^+$.

Time points are pairs $\langle i, r \rangle$ with $i \in \mathbb{N}$ (step count) and $r \in \mathbb{R}_0^+$ (time).

We say that $\langle i, r \rangle < \langle i', r' \rangle$ iff $i < i'$ or $i = i'$ and $r < r'$.

$\sigma, t \models \phi$ is defined recursively on the structure of ϕ :

$\sigma, t \models \phi_1 U \phi_2$ iff there exists $t' \geq t$, $\sigma, t' \models \phi_2$ and for all t'' , $t \leq t'' < t'$, $\sigma, t'' \models \phi_1$

$\sigma, t \models \phi_1 S \phi_2$ iff there exists $t' \leq t$, $\sigma, t' \models \phi_2$ and for all t'' , $t' < t'' \leq t$, $\sigma, t'' \models \phi_1$

$\sigma, t \models X\phi$ iff there exists $t' > t$, $\sigma, t' \models \phi$ and there exists no t'' , $t < t'' < t'$

$\sigma, t \models \tilde{X}\phi$ iff for all $t' > t$, there exists t'' , $t < t'' < t'$, $\sigma, t'' \models \phi$

$\sigma, t \models Y\phi$ iff $t > 0$ and there exists $t' < t$, $\sigma, t' \models \phi$ and
there exists no t'' , $t' < t'' < t$

$\sigma, t \models \tilde{Y}\phi$ iff $t > 0$ and for all $t' < t$, there exists t'' , $t' < t'' < t$, $\sigma, t'' \models \phi$

Usual definitions for predicates, conjunction and negation apply.

Let b be a boolean symbol. Are the following properties **true** or **false**?

$$\tilde{Y} \top :$$

$$(\neg Xb) \rightarrow (X\neg b) :$$

$$(\neg \tilde{X} b) \rightarrow (\tilde{X} \neg b) :$$

$$(X\neg b) \rightarrow (\neg Xb) :$$

$$(\tilde{X} \neg b) \rightarrow (\neg \tilde{X} b) :$$

$$(G\tilde{X}\top) \rightarrow ((Gb) \vee (G\neg b)) :$$

Let b be a boolean symbol. Are the following properties **true** or **false**?

$\tilde{Y}\top$: **false** in the initial state.

$(\neg Xb) \rightarrow (X\neg b)$:

$(\neg \tilde{X} b) \rightarrow (\tilde{X}\neg b)$:

$(X\neg b) \rightarrow (\neg Xb)$:

$(\tilde{X}\neg b) \rightarrow (\neg \tilde{X}b)$:

$(G\tilde{X}\top) \rightarrow ((Gb) \vee (G\neg b))$:

Let b be a boolean symbol. Are the following properties **true** or **false**?

$\tilde{Y}\top$: **false** in the initial state.

$(\neg Xb) \rightarrow (X\neg b)$: **false**, the first one holds in every time elapse, the second one holds only in discrete steps where $\neg b$ holds in the next state.

$(\neg \tilde{X} b) \rightarrow (\tilde{X}\neg b)$:

$(X\neg b) \rightarrow (\neg Xb)$:

$(\tilde{X}\neg b) \rightarrow (\neg \tilde{X}b)$:

$(G\tilde{X}\top) \rightarrow ((Gb) \vee (G\neg b))$:

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$(\neg \tilde{X} b) \rightarrow (\tilde{X}\neg b)$: **false**, as above but for time elapses.

$(X\neg b) \rightarrow (\neg Xb)$:

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$(\neg \tilde{X} b) \rightarrow (\tilde{X}\neg b)$: **false**, as above but for time elapses.

$(X\neg b) \rightarrow (\neg Xb)$: **true**, the first one holds iff there is a discrete step and $\neg b$ holds in the next state, hence Xb is **false**.

$(\tilde{X}\neg b) \rightarrow (\neg \tilde{X}b)$:

$(G\tilde{X}\top) \rightarrow ((Gb) \vee (G\neg b))$:

Let b be a boolean symbol. Are the following properties **true** or **false**?

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$(G\tilde{X}\top) \rightarrow ((Gb) \vee (G\neg b))$:

Let b be a boolean symbol. Are the following properties **true** or **false**?

$\tilde{Y}T$: **false** in the initial state.

$(\neg Xb) \rightarrow (X\neg b)$: **false**, the first one holds in every time elapse, the second one holds only in discrete steps where $\neg b$ holds in the next state.

$(\neg \tilde{X} b) \rightarrow (\tilde{X}\neg b)$: **false**, as above but for time elapses.

$(X\neg b) \rightarrow (\neg Xb)$: **true**, the first one holds iff there is a discrete step and $\neg b$ holds in the next state, hence Xb is **false**.

$(\tilde{X}\neg b) \rightarrow (\neg \tilde{X}b)$: **true**, as above but for time elapses.

$(G\tilde{X}T) \rightarrow ((Gb) \vee (G\neg b))$: **true**, the first part implies that we never perform a discrete transition and the truth value of b can only change in discrete transitions.



LTL- MTL Properties [2/2]

See files in examples.

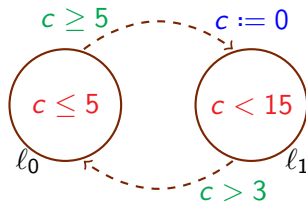


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Exercise 11.1: Simple Timed Automaton

Write the SMV model corresponding to the timed automaton in the figure.



Encode the following properties:

- ▶ from location ℓ_0 we always reach ℓ_1 within 5 time units
- ▶ if we are in ℓ_1 then for the next 3 time units we remain in ℓ_1
- ▶ if just arrived in ℓ_1 then for the next 3 time units we remain in ℓ_1 .

Exercise 11.2: Timed Thermostat

A thermostat has 2 states: **on** and **off**;

- ▶ if the temperature is **below 18** degrees the thermostat switches **on**.
- ▶ if the temperature is **above 18** degrees the thermostat switches **off**.

Every time the thermostat measures the temperature in the room, the temperature **increases** (if **on**) or **decreases** (if **off**) by dt (with respect to the previous check).

The thermostat measures the temperature at most ($\leq$) every max_dt time units.

The temperature initially is in $[18 - max_dt, 18 + max_dt]$.

Verify that the temperature is always in $[18 - 2 \cdot max_dt, 18 + 2 \cdot max_dt]$

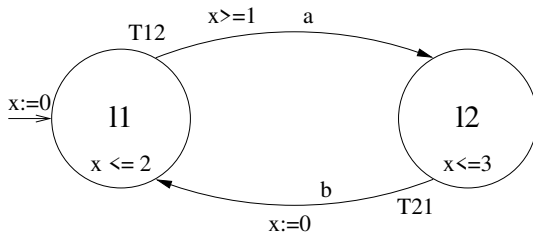


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Homework 11.1: Timed Automata

Encode the timed automata in the figure below in nuXmv.

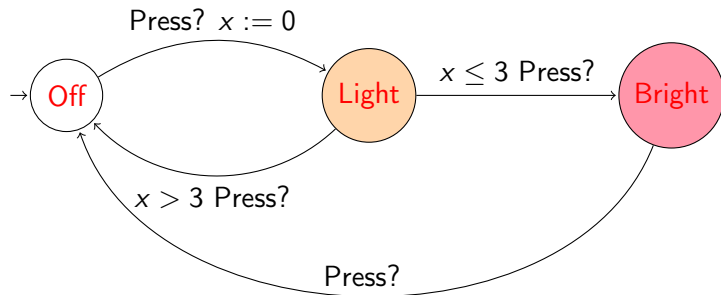


Check the following properties:

- ▶ From location ℓ_2 we always reach ℓ_1 within 1 time unit
- ▶ From location ℓ_2 we always reach ℓ_1 within 2 time units
- ▶ It is possible to be in location ℓ_1 at time $t = 1$

Homework 11.2: Light Control

Encode the timed automata in the figure below in nuXmv.



Verify the following properties:

- ▶ If the button is pressed infinitely often, then the light is turned on infinitely often.
- ▶ If the button is pressed infinitely often within 2 time units, then the light is bright infinitely often.



References

- [1] [Alessandro Cimatti et al.](#) "Extending nuXmv with Timed Transition Systems and Timed Temporal Properties". In: *CAV 2019*.